



Designation: E1001 – 21

Standard Practice for Detection and Evaluation of Discontinuities by the Immersed Pulse-Echo Ultrasonic Method Using Longitudinal Waves¹

This standard is issued under the fixed designation E1001; the number immediately following the designation indicates the year of original adoption or, in the case of revision, the year of last revision. A number in parentheses indicates the year of last reapproval. A superscript epsilon (ε) indicates an editorial change since the last revision or reapproval.

This standard has been approved for use by agencies of the U.S. Department of Defense.

1. Scope*

1.1 This practice describes procedures for the ultrasonic examination of bulk materials or parts by transmitting pulsed, longitudinal waves through a liquid couplant into the material and observing the indications of reflected waves (see Fig. 1). It covers only examinations in which one search unit is used as both transmitter and receiver (pulse-echo) and in which the part or material being examined is coupled to the part by a liquid column or is totally submerged in the couplant (either method is considered to be immersion testing). This practice includes general requirements and procedures which may be used for detecting discontinuities and for making a relative or approximate evaluation of the size of discontinuities.

1.2 This practice replaces Practice E214 and provides more detailed procedures for the selection, standardization, and operation of an examination system and for evaluation of the indications obtained.

1.3 *Units*—The values stated in inch-pound units are to be regarded as standard. The values given in parentheses are mathematical conversions to SI units that are provided for information only and are not considered standard.

1.4 *This standard does not purport to address all of the safety concerns, if any, associated with its use. It is the responsibility of the user of this standard to establish appropriate safety, health, and environmental practices and determine the applicability of regulatory limitations prior to use.*

1.5 *This international standard was developed in accordance with internationally recognized principles on standardization established in the Decision on Principles for the Development of International Standards, Guides and Recommendations issued by the World Trade Organization Technical Barriers to Trade (TBT) Committee.*

¹ This practice is under the jurisdiction of ASTM Committee E07 on Nondestructive Testing and is the direct responsibility of Subcommittee E07.06 on Ultrasonic Method.

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2. Referenced Documents

2.1 ASTM Standards:²

C1212 Practice for Fabricating Ceramic Reference Specimens Containing Seeded Voids (Withdrawn 2018)³

C1336 Practice for Fabricating Non-Oxide Ceramic Reference Specimens Containing Seeded Inclusions (Withdrawn 2018)³

E127 Practice for Fabrication and Control of Flat Bottomed Hole Ultrasonic Standard Reference Blocks

E214 Practice for Immersed Ultrasonic Testing by the Reflection Method Using Pulsed Longitudinal Waves (Withdrawn 2007)³

E317 Practice for Evaluating Performance Characteristics of Ultrasonic Pulse-Echo Testing Instruments and Systems without the Use of Electronic Measurement Instruments

E543 Specification for Agencies Performing Nondestructive Testing

E1316 Terminology for Nondestructive Examinations

2.2 ASNT Documents:⁴

SNT-TC-1A Recommended Practice for Personnel Qualification and Certification in Nondestructive Testing

ANSI/ASNT-CP-189 for Qualification and Certification of Nondestructive Testing Personnel

2.3 AIA Document:⁵

NAS-410 Certification and Qualification of Nondestructive Testing Personnel

NOTE 1—For DoD contracts, unless otherwise specified, the issues of the documents, which are DoD adopted, are those listed in the issue of the DoDISS (Department of Defense Index of Specifications Standards) cited in the solicitation.

² For referenced ASTM standards, visit the ASTM website, www.astm.org, or contact ASTM Customer Service at service@astm.org. For *Annual Book of ASTM Standards* volume information, refer to the standard's Document Summary page on the ASTM website.

³ The last approved version of this historical standard is referenced on www.astm.org.

⁴ Available from American Society for Nondestructive Testing (ASNT), P.O. Box 28518, 1711 Arlington Ln., Columbus, OH 43228-0518, <http://www.asnt.org>.

⁵ Available from Aerospace Industries Association of America, Inc. (AIA), 1000 Wilson Blvd., Suite 1700, Arlington, VA 22209-3928, <http://www.aia-aerospace.org>.

*A Summary of Changes section appears at the end of this standard

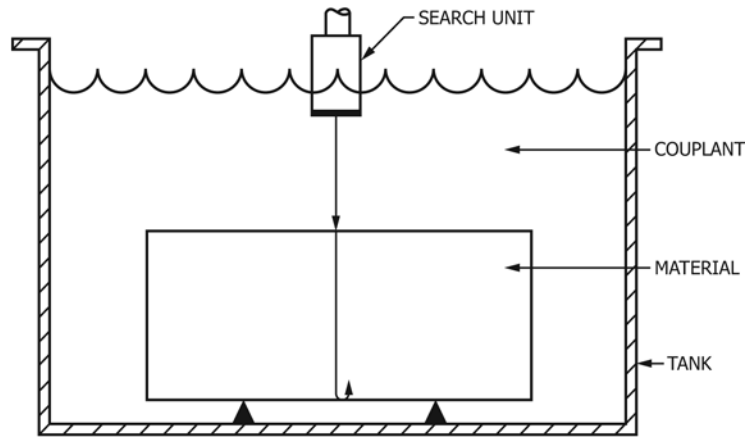


FIG. 1 Basic Immersion Setup

2.4 ISO Document:⁶

ISO 9712 Non-destructive Testing—Qualification and Certification for NDT Personnel

3. Terminology

3.1 *Definitions*—For definitions of terms used in this practice, see Terminology E1316.

3.2 *Definitions of Terms Specific to This Standard:*

3.2.1 *effective beam diameter, n*—that distance through which a search unit can be traversed across a reference reflector so that the corresponding echo amplitude is at least one half (−6 dB) of the maximum amplitude. The effective beam diameter is not a characteristic of the search unit alone, but is dependent on propagating medium, distance to the discontinuity, reflector geometry, etc.

3.2.2 *scan index, n*—the length of the step created by indexing the scan of the search unit over the part, that is continuously scanning in one direction, then stepping in the direction perpendicular to the scan or making a linear advance per rotation (pitch) for rotary scan of cylindrical parts. The allowable scan index should be correlated with the search unit effective beam diameter to ensure full coverage of the part as described in 8.2 below.

3.2.3 *transfer, n*—a change in scanning gain to compensate for differences in attenuation of the reference standard and the part or material being examined.

4. Summary of Practice

4.1 This practice describes a means for obtaining an evaluation of discontinuities in materials by immersed examination with longitudinal ultrasonic waves. Equipment, reference standards, examination and evaluation procedures, and documentation are described in detail.

5. Basis of Application

5.1 The following items are subject to contractual agreement between the parties using or referencing this practice.

5.2 *Personnel Qualification:*

5.2.1 If specified in the contractual agreement, personnel performing examinations to this practice shall be qualified in accordance with a nationally or internationally recognized NDT personnel qualification practice or standard such as ANSI/ASNT-CP-189, SNT-TC-1A, NAS-410, ISO 9712, or a similar document and certified by the employer or certifying agency, as applicable. The practice or standard used and its applicable revision shall be identified in the contractual agreement between the using parties.

5.3 *Qualification of Nondestructive Agencies*—If specified in the contractual agreement, NDT agencies shall be qualified and evaluated as described in Specification E543. The applicable edition of Specification E543 shall be specified in the contractual agreement.

5.4 *Procedures and Techniques*—The procedures and techniques to be utilized shall be as specified in the contractual agreement.

5.5 *Surface Preparation*—The pre-examination surface preparation criteria shall be in accordance with 8.1 unless otherwise specified.

5.6 *Extent of Examination*—The extent of examination shall be in accordance with 12.3, unless otherwise specified.

5.7 *Reporting Criteria/Acceptance Criteria*—Reporting criteria and acceptance criteria for the examination results shall be in accordance with 12.3, unless otherwise specified.

5.8 *Reexamination of Repaired/Reworked Items*—Reexamination of repaired/reworked items is not addressed in this practice and if required shall be specified in the contractual agreement.

6. Significance and Use

6.1 This practice provides guidelines for the application of immersed longitudinal wave examination to the detection and quantitative evaluation of discontinuities in materials.

6.2 Although not all requirements of this practice can be applied universally to all examination situations and materials, it does provide a basis for establishing contractual criteria between suppliers and purchasers of materials for performing

⁶ Available from International Organization for Standardization (ISO), ISO Central Secretariat, BIBC II, Chemin de Blandonnet 8, CP 401, 1214 Vernier, Geneva, Switzerland, <http://www.iso.org>.

immersed pulse-echo examination, and may be used as a general guide for writing detailed specifications for particular applications.

6.3 This practice is directed towards the evaluation of discontinuities detectable at normal beam incidence. If discontinuities at other orientations are of concern, alternate scanning techniques are required.

7. Apparatus

7.1 *Electronic Equipment*—The electronic equipment shall be capable of producing and processing electronic signals at frequencies in the range of search unit frequencies being used. The equipment and its display shall be capable of meeting the requirements to be completed in Table 1, as agreed upon between the supplier and the purchaser, and as measured in accordance with procedures described in Practice E317 or equivalent procedures (see Note 2). These requirements are applicable only for the frequencies required for the examination. Also, the equipment, including the search unit, shall be capable of producing echo amplitudes of at least 60 %, of full scale, with the noise level no greater than 20 %, from the appropriate reference reflector at a material distance equal to the thickness of the part to be examined. Alternatively, if these conditions can be met at one half the part thickness, the part may be examined from both sides. The instrument must have a pulser of the sufficient voltage, repetition rate, and waveshape to provide total volume coverage at the desired scanning speed.

NOTE 2—Significantly higher frequencies than those shown in Table 1 (for example, 50 MHz) may be necessary for the smaller critical flaw size of advanced ceramics.

7.2 *Voltage Regulator*—If fluctuations in line voltage cause variations exceeding ± 5 % of the vertical limit in an indication with an amplitude of one half the vertical limit, a voltage regulator should be required on the power source. This requirement is not applicable to battery-operated units.

7.3 *Search Units*—The search unit selected shall be compatible with the electronic equipment being used and with the material to be examined. The search units shall be of the immersion type. Only straight-beam (longitudinal) search units, with flat or focused acoustic lenses, shall be used. Focused or dual element search units may provide better near-surface resolution and detection of small discontinuities. Generally, round or rectangular search units are used for examination whereas round search units with symmetrical sound beam patterns are used for evaluation.

TABLE 1 Minimum Equipment Requirements (Longitudinal Wave)

Instrument Characteristics: Ultrasonic Test Frequencies 2.25, 5.0, 10.0, 15.0 MHz
Vertical limit, in. (mm), trace to peak or percent of full screen height
Upper linearity limit, in. (mm), trace to peak or percent of full screen height
Lower linearity limit, in. (mm), trace to peak or percent of full screen height
Ultrasonic sensitivity, reflector size, material distance, in. (mm)
Signal-to-noise ratio
Entry surface resolution, in. (mm)
Back surface resolution, in. (mm)
Horizontal limit, in. (mm) or percent of full screen width
Horizontal linearity range, in. (mm) or percent of full screen width

7.4 *Alarm*—For the examination of parts or material with regular shape and parallel surfaces, such as plate, machined bar stock, and forgings, an audible alarm shall be used in preference to a visual alarm, since the examination process can be accomplished at a speed which prevents reliable visual monitoring of the instrument screen. As a matter of practicality, an audible alarm should be used in conjunction with visual monitoring wherever possible. The alarm shall be adjustable to allow triggering at any commonly required level of indication amplitude and depth of material. During operation, the audible or visible signal produced by the alarm shall be easily detectable by the operator.

NOTE 3—Alarm requirements are not applicable if recording equipment is used unless otherwise specified in the contractual agreement.

7.4.1 *Alarm Gate Synchronization*—To ensure that the alarm gate tracks the examination area, the gate shall lock on the first interface pulse from the part rather than on the initial pulse from the system. Gating from the initial pulse can result in either partial loss of the examination area from the gate, or the inclusion of the back reflection and interface signal in the gated area. This will trigger the gate as would an imperfection.

7.5 *Manipulating Equipment* shall be provided to adequately support a search tube, containing the search unit, and to allow angular adjustment so that proper alignment of the sound beam to the part or material surface can be achieved. A manipulator may be attached between the search tube and search unit to provide the necessary angular adjustments. The scanning and indexing apparatus shall have sufficient structural rigidity to provide support for the manipulator and shall allow smooth, accurate positioning of the search unit. This apparatus shall permit control of the scan in accordance with 9.3.1 and control of the index in accordance with 9.2.1. Also, the scanning apparatus shall be sufficiently rigid to keep search unit backlash to within tolerances as specified in the contractual agreement. Water-path distances shall be continuously adjustable.

7.6 *Tank*—The container or tank shall permit accurate positioning of the search unit, reference blocks, and part or material to be examined in accordance with the requirements of Section 8.

7.7 *Reference Standards*—Ultrasonic reference blocks, or reference specimens, are used to standardize the ultrasonic equipment and to evaluate the indications received from discontinuities within the part. The ultrasonic characteristics of the reference standards such as attenuation, noise level, surface condition, and sound velocity, shall be similar to the material being examined. Metal reference standards should not be used for examining ceramics because of the large differences in attenuation velocity and acoustic impedance. Standardization (1) verifies that the instrument/search unit combination is performing as required, and (2) establishes a detection level for discontinuities. Reference blocks as described in Practice E127 have been used as standards for standardizing system performance, and may continue to be so used in cases where sufficient empirical evidence has shown that satisfactory examination results are obtained. However, it is more desirable in the general case to use a part identical in shape, dimensions,

and material properties to the parts to be examined.⁷ The procedures established in Practice E127 are recommended for selection of reference standard material and manufacturing ultrasonic reference block beam testing.

7.7.1 Flat Blocks—The three most commonly used sets of reference blocks are (1) area-amplitude blocks, containing blocks with the same material path and various sizes of reference reflectors; (2) distance-amplitude blocks containing blocks with one-size reference reflector at various material paths; and (3) a combination including both area-amplitude and distance-amplitude blocks in one set. These sets are described in Practice E127. However, in general their use is not recommended for system standardization (see 7.7 above) in cases where the item under test is not flat. Other types of reference blocks may be used when mutually agreed upon between the supplier and the purchaser. Practices C1212 and C1336 containing seeded voids and seeded inclusions may be used for ceramics.

7.7.2 Curved Surfaces—Reference blocks with flat surfaces should not be used for establishing gain settings for examinations on examination surfaces with radii of curvature less than about 8 in. (203 mm). For examination surfaces with radii of curvature less than 8 in. (203 mm), reference blocks shall be within 10 % of the radius curvature of the part being examined.

7.8 Reference Reflectors (Targets)—Flat-bottom holes, (FBH), or other artificial discontinuities, located either directly in the part or material, in a representative sample of the part or material, or, if previously found to yield satisfactory examination, in reference blocks, shall be used to establish the reference echo amplitude or to perform distance-amplitude correction, or both. For most examinations, the bottom surface of a suitable diameter flat-bottom hole is the common reference reflector. However, other types of artificial discontinuities (notches, side-drilled holes, etc.) may be used when mutually agreed upon between the supplier and the purchaser. Seeded voids (Practice C1212), seeded inclusions (Practice C1336), and laser-drilled holes are common reflectors for advanced ceramics.

8. General Examination Requirements

8.1 Material Condition—Perform ultrasonic examination of parts or material before machining if surface roughness and part geometry are within the tolerance specified in the contractual agreement. Surfaces may already be sufficiently free of roughness and waviness to permit a uniform examination over the required areas. When it is determined that surface roughness precludes adequate detection and evaluation of subsurface discontinuities, smooth the areas in question by machining, grinding, or other means before the examination is performed. For ceramics, care shall be taken to avoid generating surface or near-surface cracks by the smoothing operation. During examination and evaluation, ensure that the entry surface and back surface are free of loose scale, machining, or grinding particles or other loose foreign matter.

⁷ Beck, K.H., “Limitations to the Use of Reference Blocks for Periodic and Preinspection Calibration of Ultrasonic Inspection Instruments and Systems,” *Materials Evaluation*, Vol. 57, No. 3, March 1999.

8.2 Coverage—In all examinations, perform scanning to locate discontinuities that are oriented parallel with the entry surface, or that are in a plane approximately normal to the major working direction parallel to the grain flow of the part, or both. Examine areas of the part, which have not undergone significant material flow, by methods that will detect randomly oriented discontinuities. To ensure complete coverage of the material, it is necessary that the scanning spacing (index) is less than the effective beam length in the index direction at any depth in the material. Furthermore, to ensure repeatable response at the same amplitude from a given length discontinuity, it is necessary that the scan index not exceed the absolute difference between minimum discontinuity length and beam length.

8.2.1 Resolution—If entry surface resolution (based on 2:1 signal-to-noise ratio) is not sufficient to allow detection of the required reference reflectors near the examination surfaces, perform additional examinations from the opposite side. If surface roughness prevents the required resolution from being obtained, correct the problem before performing the examination. Also, for each examination direction, perform examinations from opposite sides when the maximum material travel distance is such that the minimum size reference reflector cannot be detected by examinations applied from only one side (see 7.1).

8.3 Ultrasonic Frequency—In general, the higher frequencies provide a more directive sound beam and provide better depth and lateral resolution. The lower frequencies provide better penetration and better detection of misaligned planar discontinuities. For a particular examination, select the frequency based on the material being examined, the anticipated type of discontinuities, and other examination requirements.

9. Examination (Scanning) Procedure

9.1 System Setup:

9.1.1 Tank—Immerse the part to be examined, reference standards, and search unit in a suitable tank filled with liquid couplant.

9.1.1.1 The liquid couplant shall be clean and deaerated to eliminate attenuation of the sound beam and to improve system signal-to-noise ratio.

9.1.1.2 Care shall be taken to ensure that extraneous indications caused by particulates, air bubbles, etc. in the couplant, do not interfere with the examination at the required examination sensitivity.

9.1.1.3 Corrosion inhibitors or wetting agents may be added as long as they do not affect the material properties.

9.1.1.4 Residual suspended particulate matter and air bubbles that collect on the material and search unit surfaces shall be removed.

9.1.2 Reference Standard Selection—The reference standards shall have the size and type of reference reflectors specified in the contractual agreement. A good basic set for metals is described in Table 2 and in Practice E127 for distance-amplitude reference blocks.

NOTE 4—The recommendations of paragraphs 9.1.2.1, 9.1.2.2, and 9.1.2.3, which follow are not applicable to ceramics.

TABLE 2 Distance Amplitude Reference Block-Metal Path Increments, in. (mm)

0.125 (3.2)
0.250 (6.4)
0.375 (9.5)
0.500 (12.7)
0.625 (15.9)
0.750 (19.1)
0.875 (22.2)
1.000 (25.4)
1.250 (31.8)
1.500 (38.1)
1.750 (44.5)
2.000 (50.8)
2.250 (57.2)
2.500 (63.5)
2.750 (69.9)
3.000 (76.2)
3.250 (82.6)
3.500 (88.9)
3.750 (95.3)
4.000 (101.6)
4.250 (108.0)
4.500 (114.3)
4.750 (120.7)
5.000 (127.0)
5.250 (133.4)
5.500 (139.7)
5.750 (146.1)
6.000 (152.4)

9.1.2.1 For examination performed only in the near-field portion of the sound beam, select metal paths from those in **Table 2**. The metal paths selected should be in increments so that the maximum metal path difference between reference reflectors does not exceed the requirements described in **Table 3**. This set shall include one reference block with a metal path equal to or less than the required front surface resolution, and one approximately equal to or greater than the thickness of the part (or one half the thickness, if the part is examined from both sides).

9.1.2.2 For examination performed only in the far-field portion of the sound beam, select at least three reference blocks with the following metal paths: (1) a metal path equal to or less than the required front-surface resolution; (2) a metal path approximately equal to one half the thickness of the part; and (3) a metal path approximately equal to or greater than the thickness of the part (or the required front-surface resolution, one quarter, and one half the thickness if the part is examined from both sides).

9.1.2.3 For examinations which are performed so that part of the thickness of the part is in the near field and part is in the far field, the set of reference block metal paths shall include blocks which satisfy the above near-field requirements covering the range from the front-surface resolution to the near-field limit and one reference block with a metal path equal to or

greater than the thickness of the part (or one half the thickness if the part is examined from both sides).

9.1.3 *Search Unit Adjustment*—Normalize the ultrasonic beam by adjusting the search unit for maximum echo amplitude from the front surface of the part or material. This is accomplished by angling the search unit in two directions, perpendicular to one another and parallel to the sound-entry surface (**Note 5**). This may be accomplished by use of a manipulator with two axes of motion, mutually perpendicular to one another, or in the case of curved surfaces, by use of a manipulator with one axis of motion and a search tube which can be rotated to achieve perpendicularity. During examination, monitor either the front-surface or back-surface indication. If changes in the shape of the part cause the amplitude of the monitored indication to decrease by more than 50 %, re-angle the search unit as necessary over different zones to maintain the beam normal to the examination surface.

NOTE 5—For focused search units, perform beam normalization so that the centerline of the beam is perpendicular to the material entry surface.

9.1.4 *Water Path*—The distance from the face of the search unit to the front surface of the material shall be such that the second front-surface echo does not appear before the first back-surface echo. The water path distance and search unit focal length will determine whether examination will occur in the near zone, far zone, or a combination of these. For focused search units, this distance should be such that the search unit focus is within the material at the depth required to meet front-surface resolution requirements.

NOTE 6—The permissible variation in the water path depends completely on the particular system and application (that is, flat or focused search unit, shape of beam profile, etc.). For establishing the distance-amplitude relationship and evaluating discontinuities, maintain the water path to within $\pm \frac{1}{8}$ in. (± 3.2 mm). During scanning, the maximum variation shall not exceed the amount specified in the contractual agreement or approved examination procedure.

9.2 Initial Scanning Standardization:

9.2.1 *Scan Index Determination*—Using the reference blocks selected in **9.1.2** and the search unit setup in **9.1.3**, determine the maximum allowable scan index as follows: (1) maximize the echo amplitude from the reflector in each reference block and adjust the amplitude from 50 to 100 % of the upper linearity limit; and (2) determine the total traversing distance in the index direction, across each reference target, through which no less than that percentage of the maximized amplitude is obtained, which corresponds to the allowable variation during repeated runs of the reference standard.⁸ This distance is dependent on the material travel to the reflector and will vary from one reference target to another. This is the effective beam diameter at each material distance. The least of the distances shall be used as the maximum allowable scan index.

9.2.2 *Distance-Amplitude Relationship*—The following paragraph provides a procedure to determine the distance-amplitude relationship for reference blocks described in **9.1.2**.

TABLE 3 Reference Block-Metal Path Selection in Near Field

Metal Path Range, in. (mm)	Maximum Metal Path Difference Between Adjacent Reference Blocks, in. (mm)
0 to 0.25 (0 to 6.4)	0.125 (3.2)
0.25 to 1.0 (6.4 to 25.4)	0.250 (6.4)
1.0 to 3.0 (25.4 to 76.2)	0.500 (12.7)
Over 3.0 (over 76.2)	1.000 (25.4)

⁸ Beck, K. H., "Effect of Ultrasonic Transducer Beam Profile on Accuracy of Discontinuity Detection During Scanning," *Materials Evaluation*, Vol. 64, No. 2, Feb. 2006, pp. 102–105.

This procedure is not mandatory, but is recommended when using electronic equipment lacking electronic distance–amplitude compensation.

9.2.2.1 Determine the distance–amplitude relationship for the set of reference blocks selected in 9.1.2 by positioning the search unit over each reference block to maximize the echo amplitude from the corresponding reference reflector. With the instrument controls (for example, pulse length and tuning) set to achieve the required resolution, select the reference block which provides the largest amplitude and adjust the gain to obtain an indication which is 80 % to 90 % of the upper linearity limit. Mark the amplitude of the maximized indication from each reference block on the display screen, and connect the points with a smooth curve. Once this is done, display screen time–based controls (for example, sweep delay and length) shall not be changed. A typical distance–amplitude curve for examinations in both the near and far fields is shown in Fig. 2.

NOTE 7—If a rectangular search unit is used for initial scanning, use the least sensitive portion of the effective beam width to determine the distance–amplitude curve.

9.2.3 *Scanning Gain Determination*—Determine the gain setting for initial scanning without or with electronic distance–amplitude compensation.

NOTE 8—For manual scanning, it is recommended that the initial scanning gain level be increased by 6 dB with no change in the alarm level.

9.2.3.1 *Without Electronic Distance–Amplitude Compensation*—Set the initial scanning gain by selecting the reference block with a material path which provides the lowest echo amplitude on the distance–amplitude curve as determined in 9.2.2. Maximize the amplitude from the reference reflector in this reference block, and adjust the instrument gain to obtain an amplitude equal to 80 % to 90 % of the upper linear limit. This gain setting is the initial scanning gain level.

9.2.3.2 *With Electronic Distance–Amplitude Compensation*—Electronic distance amplitude compensation generally uses either a time–varying–gain amplifier which adjusts all signals to approximately the same level or time–varying trigger level controls in the gating and alarm circuit. For systems that employ time–varying gain amplifiers, adjust the compensation controls so that the indication amplitudes of

all reference blocks selected in 9.1.2 are approximately equal and so that the lowest amplitude is 80 % to 90 % of upper linear limit. This gain is the initial scanning gain level. For systems that employ time–varying trigger level controls, select the reference target with a material path which provides the *highest* echo amplitude on the distance amplitude curve as determined in 9.2.2. Maximize the amplitude from the reference reflector in this block, and adjust the instrument gain to obtain an amplitude equal to 80 % to 90 % of the upper linear limit. This gain is the initial scanning gain level.

9.2.3.3 *Transfer* is sometimes required because the reference blocks being used do not have the same attenuation properties as the part or material being examined. Two widely used methods are described in Appendix X1. The technique used shall be mutually agreed upon between the supplier and the purchaser.

9.2.4 *Alarm Setting*: (If used, see Note 3 under 7.4.)

9.2.4.1 For systems adjusted as in 9.2.3.1 or 9.2.3.2, adjust the alarm trigger level to trigger at 50 % of the rejection limit established by any target in the reference standard. This corresponds to one half the indication amplitude of the lowest point on the distance–amplitude curve.

9.2.4.2 For systems with time–varying trigger level controls, set the compensation controls so that any indications exceeding one half of the reference value at the equivalent metal distance will trigger the alarm. Alternatively, the gain may be increased by 6 dB (doubling the amplitude) and the compensation control set to trigger the alarm at the full reference value at the equivalent metal distance.

9.2.4.3 *Back-Reflection Monitor*—If simultaneous monitoring of back reflection amplitude is desired or required, a dual gating/alarm system is necessary. One gate is set to monitor internal discontinuity indications, and the other is set to monitor back–reflection amplitude as stipulated by the contractual agreement. The operator shall determine that the trigger level is actually set at the required percent of back–reflection amplitude, not percent of vertical limit, since the true back–reflection amplitude is often greater than the vertical limit.

9.3 Initial Scanning Procedure:

9.3.1 *Scanning Speed*—The maximum scanning speed to be used on the part shall provide a clear indication of true–echo amplitude and activate the alarm as appropriate. Check this by scanning all reference blocks utilized in establishing the acceptance/rejection level. However, deviation from the above statement may be made provided that electronic or chart–type recording equipment is used, and the response time of the recording equipment is compatible with the scanning speed and other system parameters.

9.3.2 *Coverage*—The surfaces of the examination piece that will be scanned shall be as specified in the contractual agreement.

9.3.3 *Scanning*—Position the search unit over the part that will be examined using the same search unit–to–part distance (water path) and angular relationship as in setup. The gain should be as described in 9.2.3, corrected for transfer (9.2.3.3) if appropriate. A higher scanning gain may be used by adding a controlled amount of gain. Scan the part at the scanning

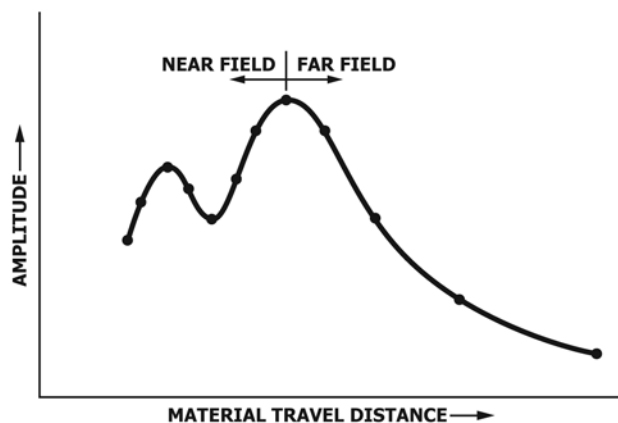


FIG. 2 Typical Distance–Amplitude Curve

speed as described in 9.3.1 (or slower) and the scan index as described in 9.2.1 (or smaller).

9.3.4 *Indications*—Note for evaluation all locations that give indication amplitudes which are greater than one half of the reference response at the equivalent depth in the material, that is that trigger the alarm.

9.3.5 *Loss of Back Reflection*—If back-reflection monitoring is required, note any location where the amplitude of the back-surface reflection is below the specified value. Determine that this loss is not caused by non-parallel surfaces or surface roughness. If surface roughness is found to be the cause of back reflection loss, the entire examination item shall be reviewed for conformance to 8.1.

10. Evaluation of Discontinuities

10.1 *Single Discontinuities*—Select the reference block with the reference reflector equal to the largest acceptable as specified in the contractual agreement and the material path distance closest to the discontinuity depth in the part, or use the applicable distance-amplitude curve established as in 9.2.2. The gain shall be as set in 9.2.3, adjusted for transfer if applicable. Manipulate the search unit to obtain the maximum echo amplitude from the discontinuity in the part. As described in 10.1.1, dynamic “evaluation” may be performed unless prohibited by contract.

10.1.1 *Accept/Reject*—If all that is required of the evaluation is an accept/reject decision, compare the amplitude of the discontinuity indication to the amplitude of the reference reflector indication. Any discontinuity from which the amplitude is greater than the reference amplitude should be marked for rejection. Unless prohibited by contract, dynamic accept/reject may also be employed, where the amplitude of the discontinuity detected during scanning is compared to the reference or reject level established at standardization; however, the amplitude repeatability of the standard reflector(s) must be established at the initial standardization.

10.1.2 *Quantitative Evaluation of Relative Discontinuity Size*—If some quantitative indication of relative discontinuity size is required, additional reference blocks, with different size reflectors, at the proper material distance are required. If the maximized amplitudes are the same, a discontinuity indication can be described as “equivalent to the response from a (specified artificial discontinuity),” for example, “equivalent to the response from a No. 5 flat-bottomed hole.” This does not mean that the two discontinuities are the same size, shape, or orientation.

NOTE 9—Size-amplitude relationships of this type are generally valid only if the ultrasonic beam is much larger than the discontinuity size. Caution should be used in evaluating discontinuities based on relative amplitude when focused, dual-element, or other highly directive search units are used.

10.2 *Linear or Multiple Discontinuities*—Evaluate linear and multiple discontinuities by first resetting the gain to achieve an echo amplitude of 80 % upper linear limit from a reference block with the reference reflector equal to the largest acceptable. Use a reference block with material travel distance closest to the discontinuity depth in the part or use the applicable distance-amplitude curve established in 9.2.2. If applicable, apply the transfer technique as described in 9.2.3.3.

10.2.1 *Multiple*—Determine the distance between discontinuities by positioning the search unit over each discontinuity where the maximum echo amplitude is obtained. Mark for rejection any part or material where the distance between the locations of maximum amplitudes of any two discontinuities is closer than the minimum allowed in the contractual agreement. It should be noted that maximum amplitude is not necessarily received from the center of a discontinuity.

10.2.2 *Linear*—Determine the approximate length of linear discontinuities having echo amplitudes which are greater than one half the amplitude of the indication from the reference reflector at the equivalent depth. Position the search unit over one extremity of the discontinuity when the amplitude is reduced to half the transfer corrected (if applicable) reference amplitude. Move the search unit toward the opposite extremity of the discontinuity until the amplitude is again reduced to half the reference. The distance between these two positions approximates the discontinuity length. Mark for rejection any part or material with linear discontinuities longer than the maximum allowed in the contractual agreement.

11. Quality Assurance Provisions

11.1 *Personnel Qualifications*—If specified in the contractual agreement, personnel performing examinations to this practice shall be qualified in accordance with one of the documents listed in 2.2 and 2.3. The practice or standard used and its applicable revision, if any, shall be identified in the contractual agreement between the using parties.

11.2 *System Performance*—As a minimum requirement, system performance shall be verified in accordance with the following schedule (if mutually agreed upon, more stringent or frequent checks may be specified): (1) Standardization should be verified after any interruption of power, change of operating personnel, replacement of a system component, or adjustment of any electrical or mechanical control which cannot be returned exactly to its previous position or value, (2) verification should also be made at such interim periods as are needed to assure repeatability of the standardization, which provides that any material previously examined can be recovered and re-examined if the standard reflector values are not within the established tolerance, and (3) at the conclusion of the inspection process (usually based upon the material lot, heat, etc.).

11.3 *Corrosion Inhibitor and Wetting Agent Control*—When corrosion inhibitor or wetting agent solutions are used in immersion tanks, check the inhibitor and agent concentration in the tank solutions after initial solution makeup and at intervals as recommended by the manufacturer. The corrosion inhibitor is used to neutralize the couplant to minimize the possibility of corrosion, rusting, pitting, etc., which can be detrimental to the part or material under examination. Wetting agents are used to deaerate the couplant and enhance adherence of the couplant to the material and search unit.

12. Documentation

12.1 Document all specific examination requirements, procedural details, and results for a particular examination in written contractual agreements, procedures, and reports.

12.2 *Contractual Agreement*—Specific examination requirements for a particular examination item shall include at least the following requirements:

- 12.2.1 Minimum equipment requirements (7.1 and Table 1),
- 12.2.2 Positioning backlash (7.5),
- 12.2.3 Reference standards (7.7.1 and 7.7.2),
- 12.2.4 Reference reflectors (7.8),
- 12.2.5 Material condition (8.1),
- 12.2.6 Water path variation (9.1.4),
- 12.2.7 Transfer technique (9.2.3.3),
- 12.2.8 Back reflection monitoring (9.2.4.3),
- 12.2.9 Coverage (9.3.2),
- 12.2.10 Evaluation of multiple discontinuities (10.2.1),
- 12.2.11 Evaluation of linear discontinuities (10.2.2),
- 12.2.12 Personnel qualification (5.1),
- 12.2.13 System performance (11.2), and
- 12.2.14 Report requirements (12.3).

12.3 *Written Procedure and Report*—Ultrasonic examinations performed in accordance with this practice shall be detailed in a written procedure. This shall identify the type of ultrasonic equipment, examination techniques, reference standards, search unit type, style, and frequency, method of reporting indications, and all other instructions that pertain to the actual examination. Procedures shall be sufficiently detailed so that another qualified operator could duplicate the examination and obtain equivalent information. All specified data required for the complete record and report of an examination shall be agreed upon between the supplier and the purchaser. As a minimum, the following items shall be documented either in the written procedure or report:

12.3.1 Specific part number and configuration examined (description of a part family is acceptable, for instance "1-2 in. thick hot-rolled stock", etc.), stage of fabrication of the part, surface finish, and surface preparation methods.

12.3.2 Manufacturer, model number, and serial numbers of all instrumentation used in the examination. This includes any recording equipment, alarm equipment, and electronic distance-amplitude equipment.

12.3.3 Type, serial number, and size of search unit. Include frequency, transducer element material (or model number), description of focal length, or search unit stand-off attachments.

12.3.4 Description of manipulating and scanning equipment, and special fixtures.

12.3.5 Couplant, corrosion inhibitor, and wetting agent solution.

12.3.6 Scanning plan. Describe the surface from which the examinations were performed and the ultrasonic beam paths used.

12.3.7 Method of applying transfer and amount of transfer applied.

12.3.8 Acceptance criteria, reporting criteria, reference standards, water paths, scan-index determination, and distance-amplitude correction.

12.3.9 Evaluation procedure.

13. Keywords

13.1 immersion ultrasonics; nondestructive testing; pulse-echo; ultrasonic examination

APPENDIXES

(Nonmandatory Information)

X1. TRANSFER TECHNIQUES

X1.1 The difficulty in transfer measurements, as they are now implemented, is in obtaining meaningful "back-surface echo" or "relative attenuation" measurements from a reference block containing an artificial discontinuity. Since a common reference block may be a cylinder containing a flat-bottom hole, the problem is described here in those terms. On axis, where the measurement is truly meaningful, the back-surface echo is reduced by scattering from the hole bottom and counterbore, if any. The amount of this reduction depends on search unit size and frequency, hole diameter, metal and water distances, whether a counterbore exists, etc. However, if the measurement is made off axis, the echo amplitude may be affected by different microstructure and sidewall reinforcement (wave guide effect) of the ultrasonic beam. The magnitude of these effects is dependent on the manufacturing process (for example, rolled or extruded) and search unit size and frequency. These two methods are described below.

X1.2 *Method A—Centerline Method* (see Fig. X1.1):

X1.2.1 The reference block shall be the one with a total length, to the back surface, approximately equal to the thickness of the part to be examined. Place the search unit over the reference block and manipulate for normal incidence, with the water path equal to that to be used in the examination. Position the search unit for a maximum flat-bottom hole echo amplitude.

X1.2.2 Without moving the search unit, adjust the instrument gain so that the first back reflection amplitude is 80 % of the upper linear limit.

NOTE X1.1—A small signal from the FBH might be present on the A-scan display but probably will not be seen at the gain setting used.

X1.2.3 Place the search unit over the material with the same water path and gain setting as above. Manipulate for maximum back reflection amplitude. Read this amplitude. The ratio of these amplitudes is a relative attenuation comparison expressed either in $\pm\%$ or $\pm\text{dB}$. If the amplitude from the part is less than 80 %, the part is more attenuating than the reference block, and

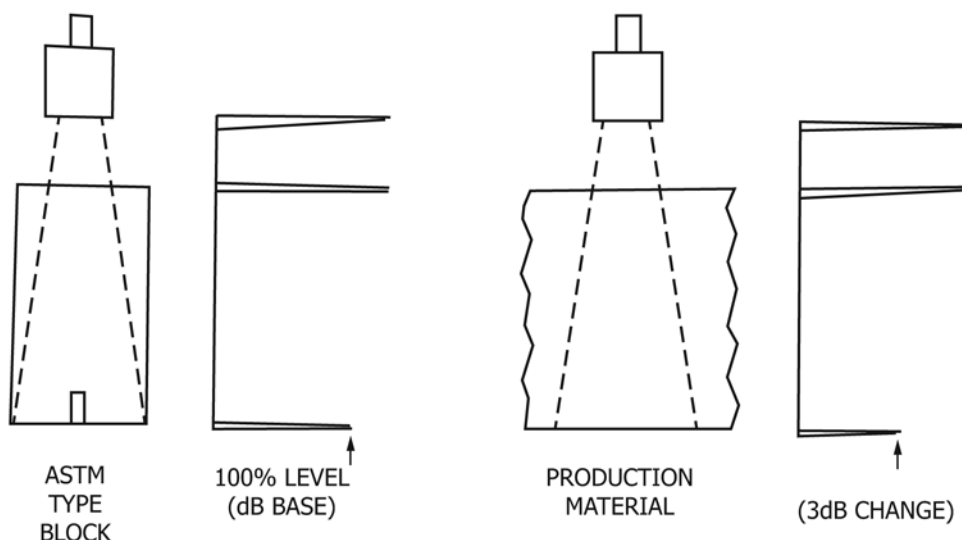


FIG. X1.1 Method A, Centerline

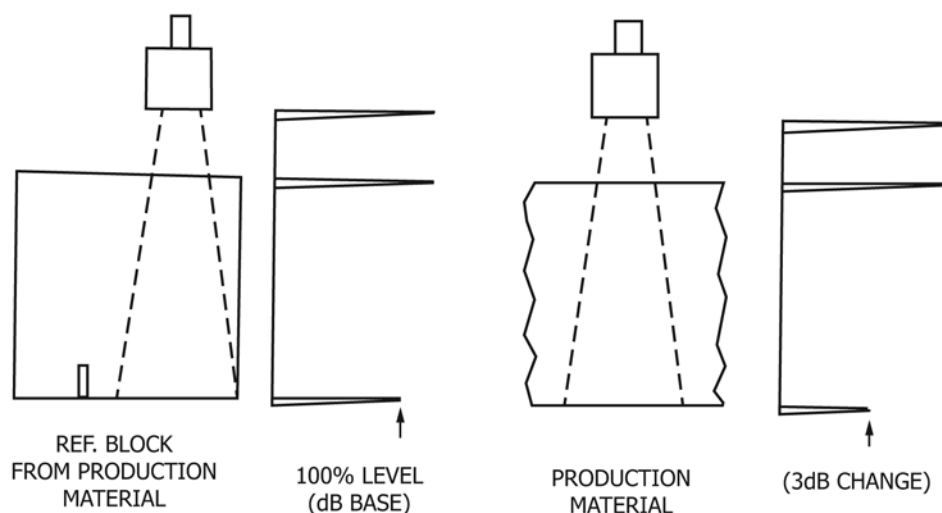


FIG. X1.2 Method B, Off Axis

unless modified by the contractual agreement, the calculated percentage or dB of gain should be added to ensure detection of discontinuities deep in the part. Transfer is not allowed if the ratio of these amplitudes is not within 25 % to 160 % or -12 dB to +4 dB.

X1.3 Method B—Off-Axis Method (see Fig. X1.2):

X1.3.1 The reference block shall be the one with a total length, to the back surface, approximately equal to the thickness of the part to be examined. The search unit shall be placed over the reference block and manipulated for normal incidence, with the water path equal to that to be used in the examination. Position the search unit off axis approximately halfway between the flat-bottom hole and the side wall. Without moving

the search unit, adjust the instrument gain so that the first back reflection amplitude is 80 % of the upper linear limit.

X1.3.2 Place the search unit over the part with the same water path and gain setting as above. Manipulate for maximum back reflection amplitude. Read this amplitude. The ratio of these amplitudes is a relative attenuation comparison, expressed either in $\pm\%$ or $\pm\text{dB}$. If the amplitude from the part is less than 80 %, the part is more attenuating than the reference block, and unless modified by the contractual agreement, the calculated percentage or dB of gain shall be added to ensure detection of discontinuities deep in the part. Transfer is not allowed if the ratio of these amplitudes is not within 25 % to 160 % or -12 dB to +4 dB.

X2. SPECIAL TECHNIQUES

X2.1 The following two techniques, while not strictly immersion in the pure sense, are very similar. Many of the procedures of this proposed recommended practice may be applied to examinations using these techniques.

X2.1.1 *Bubbler Technique*—The bubbler technique is essentially a variation of the immersion method, where the sound beam is projected through a water column into the specimen. The bubbler is usually used with an automated system for high-speed scanning of plate, sheet, strip, cylindrical forms, and other regularly shaped parts. The sound beam is projected through a column of flowing water, and is directed perpendicular to the surface for straight beam longitudinal examination. Generally, the part is not immersed (see Fig. X2.1).

X2.1.2 *Wheel Search Unit Technique*—The wheel search unit technique is an aspect of the immersion method in that the sound beam is projected through a liquid-filled tire into the

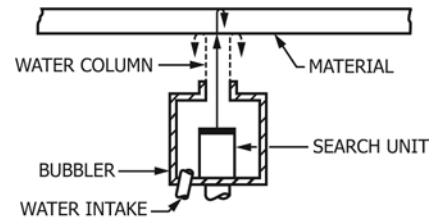


FIG. X2.1 Bubbler Setup

part. However, additional couplant is normally required between the tire and material as in contact testing. The wheel may be mounted on a stationary fixture while the material is moved past it. The position and angle of the search unit mounting on the wheel axle is variable. This method may be used for high-speed scanning of plate, sheet, strip, and other regularly shaped parts (see Fig. X2.2).

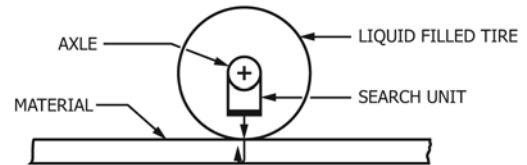


FIG. X2.2 Wheel Scanner

X3. ESTABLISHMENT OF COVERAGE BASED ON WORST-CASE INTERCEPTION

X3.1 To achieve 100 % coverage of the material under test, beam dimension must be considered. The beam may be either wider or narrower than the reference reflector or discontinuity of interest, with each case requiring attention to the scan increment in order to achieve the 100 % coverage. This consideration is essential, to avoid the condition where detection of the discontinuity of interest is dependent upon the starting point of the scan itself. Ken Beck⁸ has termed the condition where the discontinuity can be detected with any starting condition as “invariant worst-case interception.”

X3.2 It is possible to perform a scan which results depend upon the starting point on the product (and in the case of cylindrical product, the rotational position of the material under test). Therefore, it is essential to take into consideration the relationships between beam width, reflector length, and the scanning pitch or index. These relationships follow:

$$I_{wc} = (N + B - P)/2 \quad (X3.1)$$

where:

- I_{wc} = the length of the worst case beam intersection of a linear reflector,
- N = reflector length,
- B = beam width, and
- P = the scan increment.

To avoid varying response based upon the starting point of the scan, one of the following applies:

$$I_{wc}/B = 1 \text{ for } B < N, \text{ or} \quad (X3.2)$$

$$I_{wc}/N = 1 \text{ for } B > N \quad (X3.3)$$

From the above two conditions, the relationship between the pitch and the beam width is established:

$$P < |N - B| \quad (X3.4)$$

X3.3 The essential requirement is that the scan increment must be less than the width of the beam in order to provide overlap, or a minimum of 100 % coverage. This overlap is often described as a percentage in specifications, for instance a 10 % or 20 % beam overlap.

X3.4 The beam shape must also be considered. A varying beam shape will cause varying received signal amplitudes – the reflector will be detected, but the amplitude of the signal received will not be consistent, and can result in inappropriate disposition of the discontinuity in the part under test. In the worst case, a rejectable discontinuity could be accepted even after achieving a full coverage scan. In general, the instrument gain or gate settings for reliable disposition of discontinuities will be empirically determined, based upon the specific part being scanned, the instrumentation, scanning apparatus, and beam profile of the search unit. These determinations must be made at the full scanning speed of the system used for inspection of material, which will also then include the effects of product straightness/flatness, system or part vibration, and other factors present in the dynamic scanning process.

Frequently, rejection thresholds of 50 % of the dynamic standardization threshold are required, with the precise level determined by specification requirements or the user's empirical assessment.

SUMMARY OF CHANGES

Committee E07 has identified the location of selected changes to this standard since the last issue (E1001 – 16) that may impact the use of this standard.

- (1) Removed reference to Practice E428, since it has been withdrawn; all subsequent references to Practice E428 removed.
- (2) Modified **Table 1**, title/description, for clarity.
- (3) Modified description of manipulation in **7.5**, since on cylindrical products a “gimbal/swivel” type manipulation can provide the required normalization.
- (4) Removed “advanced” in the description of ceramics in **7.7** (also in **8.1** and **Note 4**). Changed “much” to “sufficient”—“much” cannot be defined, “sufficient” may be demonstrated.
- (5) Clarified that flat standard sets are less than optimal for product which has a curved or non-flat surface in **7.7.1**.
- (6) Modified the units to reflect the same conversion of 8 in. to 203 mm in **7.7.2**.
- (7) Moved the discussion of invariate worst case interception from **8.2** to **Appendix X3**, to provide a more detailed discussion of the concept.
- (8) Clarified the probe manipulation process, and that other alternatives to “two mutually perpendicular planes” exist for cylindrical product, in **9.1.3**.
- (9) Made an editorial modification of the statement regarding consensus on transfer methods in **9.2.3.3**.

- (10) Modified **9.3.1** to reflect the more common use of electronic recording, as opposed to paper strip charts.
- (11) Modified **10.1** and **10.2** to allow dynamic acceptance or rejection decisions.
- (12) Modified **11.2** for clarity, and to require an “ending” standardization verification in addition to initial and intermediate verifications.
- (13) Modified **11.3** to indicate that the mixture and checks should be performed in accordance with manufacturer's recommendations.
- (14) **12.3.1** now allows the procedure to address part “families” and not be restricted to specific part numbers, allowing for broader application of a given procedure (for instance, cylindrical product 2 in.-4 in. in diameter, rather than a listing of part numbers).
- (15) Moved the discussion of “Invariate Worst Case Interception” to the new **Appendix X3**, to provide a more detailed discussion of the concept without requiring the user to refer to another, non-ASTM document. That document (a paper by Ken Beck) is referenced in the event that the user would want to further explore the concept.

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